

Suggestions for PHYSICS IV TEE 2023 -2024

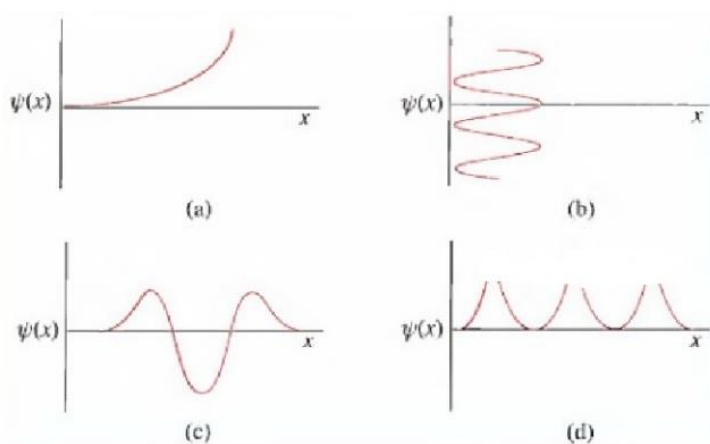
Unit 1

1. State Stefan-Boltzmann law of blackbody radiation with proper mathematical expression.
2. State Wein's displacement law of blackbody radiation with proper mathematical expression.
3. What is photoelectric effect? Summarize the failures of classical electromagnetic theory in explaining the photoelectric effect.
4. What is de Broglie hypothesis? Explain an experiment proving de Broglie hypothesis.
5. What is Compton effect?
6. State and explain Heisenberg's uncertainty principle regarding the position and momentum of a particle.
7. Explain *ultra-violet catastrophe*.
8. An enclosure, filled with *He*, is heated to 400 K . A beam of *He* atom emerges out of the enclosure. Calculate the de Broglie wavelength corresponding to *He* atoms. Mass of *He* atom is $6.7 \times 10^{-27}\text{ kg}$.
9. Light of wavelength $2000\text{ \AA}$ falls on a metal surface with work function 4.2 eV . Calculate (i) threshold wavelength and (ii) stopping potential.
10. An electron is confined in a box of length $1\text{ \AA}$. Calculate the minimum uncertainty in its velocity.
11. Can we observe the energy states of a ball of mass 100 g moving in a box of length 1 m ?
12. Calculate the de Broglie wavelength of the electron of a hydrogen atom having an energy of 13.6 eV .
13. X-ray photon of wavelength $0.3\text{ \AA}$ is scattered through an angle 45° by a loosely bound electron. Find the wavelength of scattered photon.
14. Using Heisenberg's uncertainty principle prove that electrons can't exist inside a nucleus.

Unit 2

1. What is the significance of a quantum mechanical wave function?
2. Write the quantum mechanical operators for position, momentum, total energy and the potential energy.
3. Show that the energy level of a quantum particle in an infinite potential well is quantized.
4. Differentiate between finite and infinite potential wells of quantum mechanics.
5. What is meant by "normalization" of a wave function?

6. Which of the following wavefunctions are well behaved? Give reasons for your answer.



7. An electron is trapped in a 1D region of length 10^{-10} m. How much energy must be supplied to excite the electron from the ground state to the second excited state?

8. An electron is confined to a one-dimensional potential well of length 5 Å. Find the quantized energy values for the three lowest energy states.

9. The wave function for a quantum particle of mass m confined to move in the domain $0 \leq x \leq L$ is given by

$$\psi(x) = N \sin\left(\frac{4\pi x}{L}\right)$$

where N is the normalization factor.

Normalize the wave function and write the normalized form.

10. The momentum operator is defined as $\hat{p} = -i\hbar \frac{\partial}{\partial x}$. Determine which of the following functions are the eigen functions of the momentum operator: a. $\sin kx$, b. e^{ikx} , and c. x^4 .

Unit 3

1. Explain the phenomenon of quantum tunnelling. State the various factors which determines the tunnelling probability.

2. Write a short note on Scanning Tunnelling Microscope (STM).

3. Briefly explain the decay of α - particles by applying quantum mechanical tunnelling.

4. Calculate the probability of transmission of an electron with an energy of 2 eV incident upon a rectangular potential energy barrier of height 5 eV and width 0.4 nm.

5. Write the expression for zero-point energy of a quantum harmonic oscillator and comment on the significance it carries.

Unit 4

1. When a fair coin is tossed there is an equal chance of getting either heads or tails. When two fair coins are tossed, calculate the probability that both of them will show heads.
2. When a fair dice is rolled, there is an equal chance of getting any one of its six faces. When three fair dice are rolled, calculate the probability that all will show one.
3. Explain phase space and phase point in relation with statistical physics.
4. Explain macro states and micro states in relation with statistical physics.
5. What is an ensemble? Discuss different types of ensemble in statistical physics.
6. Differentiate between MB, BE and FD statistics.
7. A system has non-degenerate single-particle states with $0, 1, 2, 3$ energy units. Three particles are to be distributed in these states such that the total energy of the system is 3 units. Find the number of microstates if the particles obey BE and FD statistics.
8. Two particles are to be distributed in two non-degenerate energy states. Find the number of distributions according to MB, BE and FD statistics.

Unit 5

1. What is the first law of thermodynamics?
2. What is the second law of thermodynamics?
3. Define system, surrounding, isolated system, closed system, open system, adiabatic process and thermal equilibrium in relation with thermodynamics.
4. Explain Carnot's cycle with appropriate diagram.
5. 2 kg of a gas is squeezed in a piston-cylinder arrangement from 14 m^3 to 9 m^3 . During this operation the pressure is kept constant as 2000 Nm^{-2} . If the internal energy of the system is decreased by 6000 J , calculate how much energy was transferred as heat to and from the gas in this process.
6. One mole of an ideal gas (obeying $PV = nRT$) is expanded isothermally at 27°C and reversibly – the initial and final volume being 2.28 m^3 and 4.56 m^3 . If the internal energy of the ideal gas is a function of temperature only, calculate the heat lost or gained by the system. [Given: gas constant $R = 8.314\text{ JK}^{-1}\text{mol}^{-1}$]
7. A Carnot engine whose heat sinks at 27°C , has an efficiency of 25% . By how many degrees should the temperature of the source be changed to increase the efficiency by 100% of the original efficiency?